GROUP 12 PRESENTATION
CASE STUDY

1. List of Entities

Department

Staff

Doctor

Nurse

AdministrativeStaff

Patient

Visit

Diagnosis

Medication

Room

Admission

2. Attributes and Keys

Department

Primary Key: DepartmentID

Attributes:

DepartmentID (PK)

DepartmentName

OfficeLocation

Staff (Superclass)

Primary Key: EmployeeID

Attributes:

EmployeeID (PK)

FullName

Address

PhoneNumbers (Multivalued)

DateOfEmployment

Doctor (Subclass of Staff)

Attributes:

EmployeeID (PK, FK)

MedicalSpecialty

Certifications (Multivalued)

Nurse (Subclass of Staff)

Attributes:

EmployeeID (PK, FK)

QualificationLevel

AdministrativeStaff (Subclass of Staff)

Attributes:

EmployeeID (PK, FK)

JobTitle

Patient

Primary Key: PatientID

Attributes:

PatientID (PK)

FullName

DateOfBirth

Gender

Address

ContactNumbers (Multivalued)

Visit

Primary Key: VisitID

Attributes:

VisitID (PK)

VisitDate

ReasonForVisit

Diagnosis

Primary Key: DiagnosisID

Attributes:

DiagnosisID (PK)

DiagnosisDescription

Medication

Primary Key: MedicationCode

Attributes:

MedicationCode (PK)

MedicationName

Manufacturer

Room

Primary Key: RoomNumber

Attributes:

RoomNumber (PK)

Admission

Primary Key: AdmissionID

Attributes:

AdmissionID (PK)

AdmissionDate

DischargeDate

3. List of Relationships

Department employs Staff

Department is managed by Doctor

Patient has_visit Visit

Doctor attends Visit

Visit results in Diagnosis

Visit prescribes Medication

Patient admitted Admission

Admission is assigned to Room

Room belongs to Department

4. Identification of Specialization / Generalization

Superclass:

Staff

Subclasses:

Doctor

Nurse

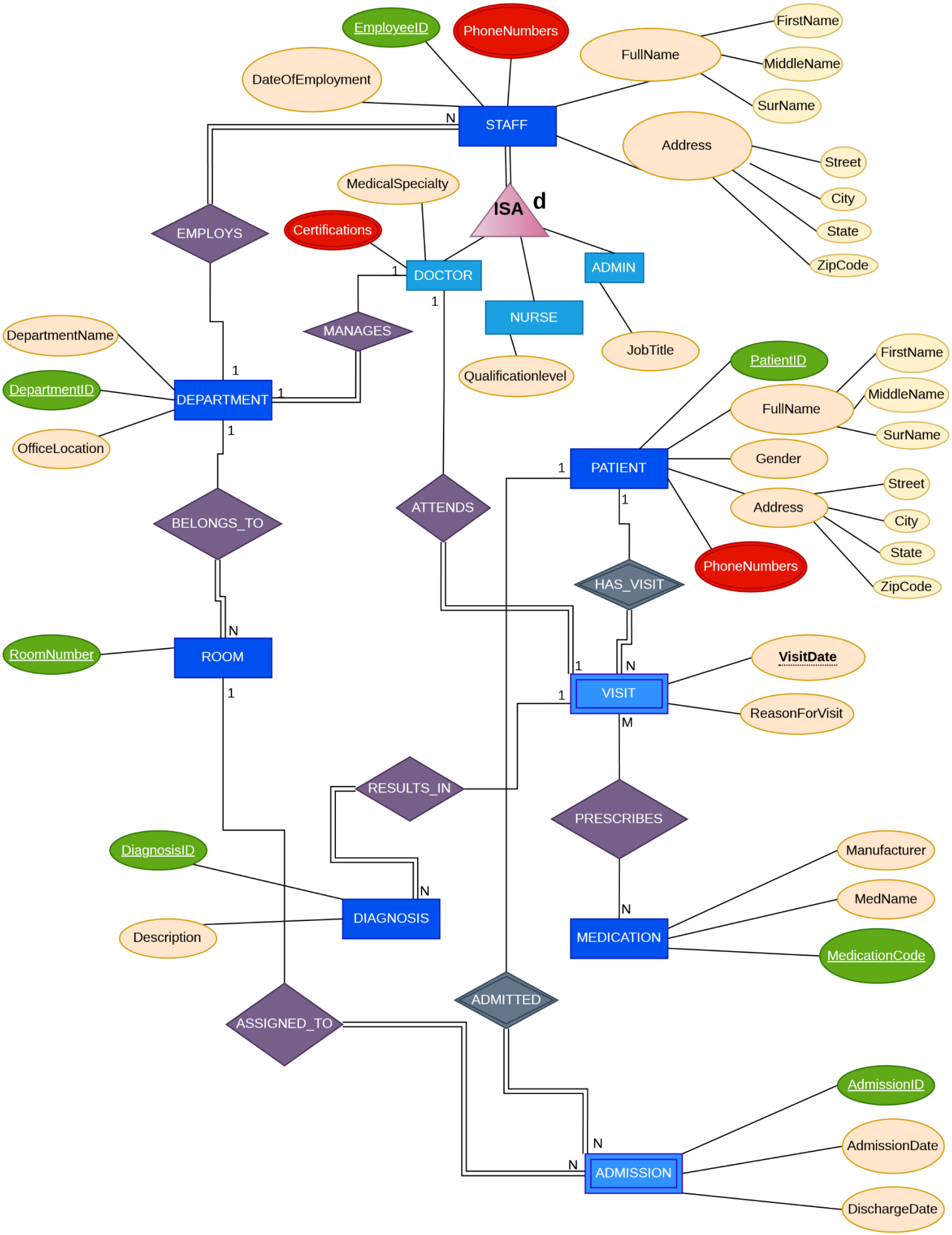
AdministrativeStaff

Type of specialization:

Disjoint specialization
(A staff member can only belong to one subclass)

Participation:

Total specialization
(Every staff member must be either a doctor, nurse, or administrative staff)


EmployeeID
PhoneNumbers
FullName
FirstName
MiddleName
SurName
DateOfEmployment
STAFF
Address
Street
City
State
ZipCode
MedicalSpecialty
ISA
d
EMPLOYS
Certifications
DOCTOR
ADMIN
NURSE
JobTitle
Qualificationlevel
MANAGES
DepartmentName
DepartmentID
DEPARTMENT
OfficeLocation
PatientID
FullName
FirstName
MiddleName
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Address
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ZipCode
PhoneNumbers
BELONGS_TO
ATTENDS
HAS_VISIT
RoomNumber
ROOM
VisitDate
VISIT
ReasonForVisit
RESULTS_IN
PRESCRIBES
DiagnosisID
DIAGNOSIS
Description
Manufacturer
MedName
MEDICATION
MedicationCode
ASSIGNED_TO
ADMITTED
AdmissionID
AdmissionDate
ADMISSION
DischargeDate
N
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M